

MOMINA WAQAR

AI/ML Engineer · Computer Vision · Automation Systems · Data Pipelines

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Summary

AI/ML Engineer who ships production AI for end-to-end systems. I train and evaluate vision models and build LLM applications. I take work from client scoping to delivery. My focus is honest evaluation: finding where a model fails, fixing it, and reporting results clients can trust.

Experience

Associate AI/ML Engineer

Apr 2026 – Present

MindVista

Remote

- **Computer vision:** trained image segmentation models (YOLO) to detect vehicle damage from photos, and evaluated DINOv2-based classifiers to measure how reliably they identify damage.
- **Data pipelines:** built and maintained Airflow ETL pipelines that onboard prediction-market and other financial datasets onto a client's data platform.
- **Document intelligence:** developed an AI tool that reviews civil-engineering plans against regulatory checklists and produces review reports, powered by Claude.
- **Document automation:** built Claude-based workflows that process client documents, handle Excel calculations and generate reports automatically.
- **Multi-agent RAG:** developed a multi-agent question-answering system over a client's internal finance documents.
- **Client communication:** worked directly with clients on requirements, proposals, demos and progress updates.

Projects

Silsila: Agent-Based Smart Redistribution System (*Final Year Project*)

FastAPI, React, PostgreSQL, Mesa

- Full-stack system that cuts waste of perishable FMCG goods. Prophet demand forecasting and a **Mesa multi-agent** engine flag expiry-risk stock and generate constraint-based redistribution plans, with a B2C clearance marketplace. Normalised PostgreSQL data model.

BridgeRoman: Urdu Voice Journal with an AI Agent

Whisper, Llama 3.3, LangChain/LangGraph, FAISS

- Low-latency Urdu speech-to-text (Whisper large-v3 via Groq) with LLM transliteration into Roman Urdu, FAISS semantic search, and a LangGraph ReAct agent that searches, summarises and translates entries.

Document-to-Markdown VLM Fine-tuning

Qwen2-VL-2B, QLoRA/PEFT, Hugging Face

- Fine-tuned a vision-language model with 4-bit QLoRA to transcribe scanned pages into Markdown/LaTeX. Reached **ROUGE-L 0.66**, with a Gradio demo.

Book Nook: AI Bookshelf (*live*)

React, TypeScript, Supabase (RLS), Groq

- Bookshelf app with Row Level Security for per-user data isolation and streaming LLM book recommendations, deployed on Vercel.

Technical Skills

Languages	Python, JavaScript/TypeScript, SQL, C++
ML & Computer Vision	PyTorch, Ultralytics YOLO (segmentation), DINOv2/ViT, OpenCV, MediaPipe, TensorFlow, scikit-learn, Prophet
LLMs & GenAI	Claude API, Retrieval-Augmented Generation (RAG), multi-agent systems, LangChain/LangGraph, fine-tuning (LoRA/QLoRA/PEFT), Hugging Face, prompt engineering, Whisper
Evaluation	mAP/IoU, AUC, ROUGE/BLEU, failure-mode audits, confusion analysis, threshold calibration
Data & MLOps	Apache Airflow, ETL, FAISS/vector search, PostgreSQL, Supabase, MongoDB, Docker, RunPod (GPU), ONNX, GCP, Azure, AWS
Backend & Web	FastAPI, Node.js/Express, React, REST APIs, Postman, Git, Vercel

Education

FAST National University of Computer and Emerging Sciences (FAST-NUCES)

Graduated June 2026

Bachelor of Science in Computer Science · CGPA: 3.2/4.00 · Dean's List (2 semesters)